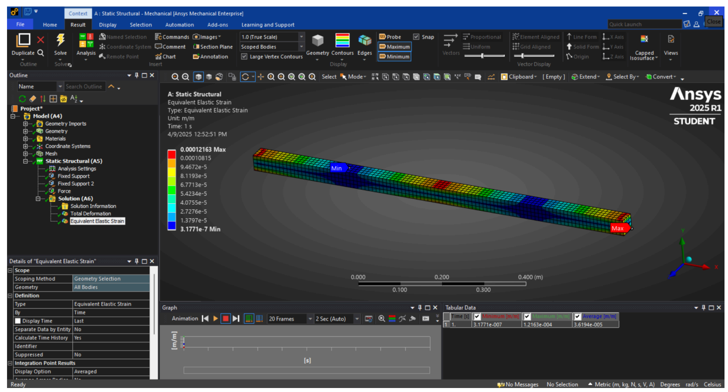
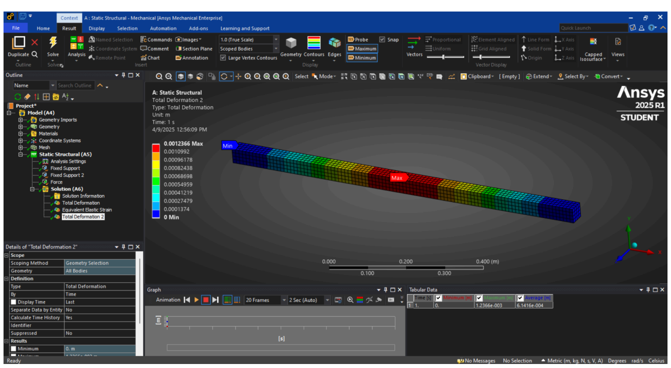
In my first ANSYS model, I performed a static structural analysis on a simply supported beam subjected to a central downward force. A key outcome of the simulation was the distribution of Von Mises stress across the beam. Von Mises stress is a scalar stress value derived from the stress tensor, which allows us to predict the yielding of materials under complex loading. It is especially useful in ductile materials like metals, where failure is often governed not by maximum normal stress but by a combination of shear and normal stresses. This makes it highly relevant for structural safety assessments in FEA simulations.



I included a strain plot in the ANSYS simulation results. Specifically, I selected the Equivalent (Von Mises) Strain from the strain options, which, like Von Mises stress, provides a scalar value that accounts for the combined effects of different strain components. When I plotted the strain, it confirmed that the beam experienced very little actual deformation under the 2 kN load. The initially exaggerated deformation shown by ANSYS was clarified by switching to “True Scale.” The Equivalent Strain plot showed low values throughout the structure, with slightly higher strain at the center, consistent with where the stress was greatest.



To explore how increased loading affects beam deformation, I modified the model by changing the applied force from 2 kN to 10 kN. After rerunning the simulation with the increased load, I reviewed the Total Deformation result. As anticipated, the deformation scaled with the increased force. The maximum deformation at 10 kN was 0.0012366 meters, compared to 0.00024731 meters at the original 2 kN load. This result showed a roughly fivefold increase in deformation, which aligns with expectations for a linear elastic material, where deformation is directly proportional to the applied load within the elastic range.

My first experience using ANSYS for finite element modeling was both insightful and challenging. Overall, this first simulation gave me a strong foundation to build on, and I'm excited to continue learning more about finite element analysis.